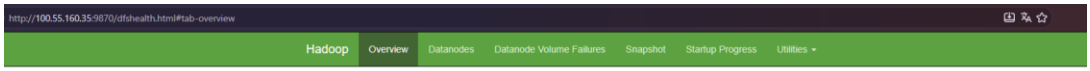
```
hadoop@ip-172-31-32-84:~$ jps
1491 NameNode
1847 SecondaryNameNode
1960 Jps
1625 DataNode
```

```
hadoop@ip-172-31-32-84:~$ hdfs dfs -mkdir -p /user/alumno/test
```

```
hadoop@ip-172-31-32-84:~$ hdfs dfs -put prueba.txt /user/alumno/test/
hadoop@ip-172-31-32-84:~$
```

```
hadoop@ip-172-31-32-84:~$ hdfs dfs -ls /user/alumno/test/
Found 1 items
-rw-r--r--   1 hadoop supergroup          4 2026-10-25 18:21 /user/alumno/test/prueba.txt
hadoop@ip-172-31-32-84:~$
```

```
hadoop@ip-172-31-32-84:~$ hdfs dfs -cat /user/alumno/test/prueba.txt
asd
hadoop@ip-172-31-32-84:~$
```



Overview 'localhost:8020' (active)

Started:	Sun Oct 25 19:00:43 +0100 2026
Version:	3.4.1, r4d7825309348956336b8f06a08322b78422849b1
Compiled:	Wed Oct 09 16:57:00 +0200 2024 by mthakur from branch-3.4.1
Cluster ID:	CID-fa2644b0-1f18-467f-a3ae-bd988a2335a0
Block Pool ID:	BP-1985927739-172.31.32.84-1792487596514

Summary

Security is off.

Safemode is off.

6 files and directories, 2 blocks (2 replicated blocks, 0 erasure coded block groups) = 8 total filesystem object(s).

Heap Memory used 62.05 MB of 230 MB Heap Memory. Max Heap Memory is 869.5 MB.

Non Heap Memory used 61.81 MB of 63.31 MB Commited Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:	6.61 GB
Configured Remote Capacity:	0 B
DFS Used:	56 KB (0%)
Non DFS Used:	4.76 GB
DFS Remaining:	1.83 GB (27.69%)
Block Pool Used:	56 KB (0%)
DataNodes usages% (Min/Median/Max/stdDev):	0.00% / 0.00% / 0.00% / 0.00%
Live Nodes	1 (Decommissioned: 0, In Maintenance: 0)
Dead Nodes	0 (Decommissioned: 0, In Maintenance: 0)
Decommissioning Nodes	0
Entering Maintenance Nodes	0
Total Datanode Volume Failures	0 (0 B)
Number of Under-Replicated Blocks	0
Number of Blocks Pending Deletion (including replicas)	0
Block Deletion Start Time	Sun Oct 25 19:00:43 +0100 2026
Last Checkpoint Time	Sun Oct 25 19:00:44 +0100 2026
Last HA Transition Time	Never
Enabled Erasure Coding Policies	RS-6-3-1024k

```
hadoop@ip-172-31-32-84:/opt/hadoop/share/hadoop/mapreduce$ hadoop jar ./hadoop-mapreduce-examples-3.4.1.jar pi 2 100
Number of Maps  = 2
Samples per Map = 100
Wrote input for Map #0
Wrote input for Map #1
Starting Job
2026-10-25 18:33:50,986 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-10-25 18:33:51,040 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-10-25 18:33:51,040 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-10-25 18:33:51,219 INFO input.FileInputFormat: Total input files to process : 2
2026-10-25 18:33:51,228 INFO mapreduce.JobSubmitter: number of splits:2
2026-10-25 18:33:51,348 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1831276806_0001
2026-10-25 18:33:51,348 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-10-25 18:33:51,489 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-10-25 18:33:51,490 INFO mapreduce.Job: Running job: job_local1831276806_0001
2026-10-25 18:33:51,493 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-10-25 18:33:51,502 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-10-25 18:33:51,503 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-10-25 18:33:51,503 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-10-25 18:33:51,506 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2026-10-25 18:33:51,545 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-10-25 18:33:51,546 INFO mapred.LocalJobRunner: Starting task: attempt_local1831276806_0001_m_000000_0
2026-10-25 18:33:51,567 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-10-25 18:33:51,567 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-10-25 18:33:51,567 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-10-25 18:33:51,589 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-10-25 18:33:51,592 INFO mapred.MapTask: Processing split: hdfs://localhost:8020/user/hadoop/QuasiMonteCarlo_1792953229317_959697275/in/part0:0+118
2026-10-25 18:33:51,611 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-10-25 18:33:51,611 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-10-25 18:33:51,611 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-10-25 18:33:51,611 INFO mapred.MapTask: soft limit at 83886080
2026-10-25 18:33:51,611 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-10-25 18:33:51,611 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-10-25 18:33:51,613 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-10-25 18:33:51,662 INFO mapred.LocalJobRunner:
2026-10-25 18:33:51,663 INFO mapred.MapTask: Starting flush of map output
2026-10-25 18:33:51,664 INFO mapred.MapTask: Spilling map output
2026-10-25 18:33:51,664 INFO mapred.MapTask: bufstart = 0; bufend = 18; bufvoid = 104857600
2026-10-25 18:33:51,664 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214392(104857568); length = 5/6553600
2026-10-25 18:33:51,671 INFO mapred.MapTask: Finished spill 0
2026-10-25 18:33:51,682 INFO mapred.Task: Task:attempt_local1831276806_0001_m_000000_0 is done. And is in the process of committing
2026-10-25 18:33:51,685 INFO mapred.LocalJobRunner: map
2026-10-25 18:33:51,685 INFO mapred.Task: Task 'attempt_local1831276806_0001_m_000000_0' done.
2026-10-25 18:33:51,692 INFO mapred.Task: Final Counters for attempt_local1831276806_0001_m_000000_0: Counters: 23
        File System Counters
                FILE: Number of bytes read=282014
                FILE: Number of bytes written=1001429
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=118
                HDFS: Number of bytes written=236
                HDFS: Number of read operations=7
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=1
                Map output records=2
                Map output bytes=18
                Map output materialized bytes=28
                Input split bytes=147
                Combine input records=0
                Spilled Records=2
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=245891072
        File Input Format Counters
                Bytes Read=118
2026-10-25 18:33:51,692 INFO mapred.LocalJobRunner: Finishing task: attempt_local1831276806_0001_m_000000_0
2026-10-25 18:33:51,693 INFO mapred.LocalJobRunner: Starting task: attempt_local1831276806_0001_m_000001_0
2026-10-25 18:33:51,694 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-10-25 18:33:51,694 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-10-25 18:33:51,694 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-10-25 18:33:51,694 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-10-25 18:33:51,695 INFO mapred.MapTask: Processing split: hdfs://localhost:8020/user/hadoop/QuasiMonteCarlo_1792953229317_959697275/in/part1:0+118
2026-10-25 18:33:51,766 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-10-25 18:33:51,766 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-10-25 18:33:51,766 INFO mapred.MapTask: soft limit at 83886080
2026-10-25 18:33:51,766 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-10-25 18:33:51,766 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-10-25 18:33:51,767 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-10-25 18:33:51,775 INFO mapred.LocalJobRunner:
2026-10-25 18:33:51,775 INFO mapred.MapTask: Starting flush of map output
2026-10-25 18:33:51,775 INFO mapred.MapTask: Spilling map output
2026-10-25 18:33:51,776 INFO mapred.MapTask: bufstart = 0; bufend = 18; bufvoid = 104857600
2026-10-25 18:33:51,776 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214392(104857568); length = 5/6553600
2026-10-25 18:33:51,776 INFO mapred.MapTask: Finished spill 0
2026-10-25 18:33:51,780 INFO mapred.Task: Task:attempt_local1831276806_0001_m_000001_0 is done. And is in the process of committing
2026-10-25 18:33:51,785 INFO mapred.LocalJobRunner: map
2026-10-25 18:33:51,785 INFO mapred.Task: Task 'attempt_local1831276806_0001_m_000001_0' done.
2026-10-25 18:33:51,786 INFO mapred.Task: Final Counters for attempt_local1831276806_0001_m_000001_0: Counters: 23
        File System Counters
                FILE: Number of bytes read=282327
                FILE: Number of bytes written=1001489
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=236
                HDFS: Number of bytes written=236
                HDFS: Number of read operations=10
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=1
                Map output records=2
                Map output bytes=18
                Map output materialized bytes=28
                Input split bytes=147
                Combine input records=0
                Spilled Records=2
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=351272960
        File Input Format Counters
                Bytes Read=118
2026-10-25 18:33:51,786 INFO mapred.LocalJobRunner: Finishing task: attempt_local1831276806_0001_m_000001_0
2026-10-25 18:33:51,787 INFO mapred.LocalJobRunner: map task executor complete.
```

```
2026-10-25 18:33:51,791 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-10-25 18:33:51,791 INFO mapred.LocalJobRunner: Starting task: attempt_local1831276806_0001_r_000000_0
2026-10-25 18:33:51,798 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-10-25 18:33:51,798 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-10-25 18:33:51,798 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-10-25 18:33:51,799 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-10-25 18:33:51,804 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle@947037d
2026-10-25 18:33:51,806 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-10-25 18:33:51,822 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit=638215744, maxSingleShuffleLimit=159553936, mergeThreshold=421222400, ioSortFactor=10, memToMemMergeOutputsThreshold=10
2026-10-25 18:33:51,825 INFO reduce.EventFetcher: attempt_local1831276806_0001_r_000000_0 Thread started: EventFetcher for fetching Map Completion Events
2026-10-25 18:33:51,857 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt_local1831276806_0001_m_000001_0 decomp: 24 len: 28 to MEMORY
2026-10-25 18:33:51,861 INFO reduce.InMemoryMapOutput: Read 24 bytes from map-output for attempt_local1831276806_0001_m_000001_0
2026-10-25 18:33:51,862 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 24, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory ->24
2026-10-25 18:33:51,865 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt_local1831276806_0001_m_000000_0 decomp: 24 len: 28 to MEMORY
2026-10-25 18:33:51,866 INFO reduce.InMemoryMapOutput: Read 24 bytes from map-output for attempt_local1831276806_0001_m_000000_0
2026-10-25 18:33:51,866 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 24, inMemoryMapOutputs.size() -> 2, commitMemory -> 24, usedMemory ->48
2026-10-25 18:33:51,867 INFO reduce.EventFetcher: EventFetcher is interrupted.. Returning
2026-10-25 18:33:51,869 INFO mapred.LocalJobRunner: 2 / 2 copied.
2026-10-25 18:33:51,869 INFO reduce.MergeManagerImpl: finalMerge called with 2 in-memory map-outputs and 0 on-disk map-outputs
2026-10-25 18:33:51,874 INFO mapred.Merger: Merging 2 sorted segments
2026-10-25 18:33:51,875 INFO mapred.Merger: Down to the last merge-pass, with 2 segments left of total size: 42 bytes
2026-10-25 18:33:51,875 INFO reduce.MergeManagerImpl: Merged 2 segments, 48 bytes to disk to satisfy reduce memory limit
2026-10-25 18:33:51,876 INFO reduce.MergeManagerImpl: Merging 1 files, 50 bytes from disk
2026-10-25 18:33:51,876 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from memory into reduce
2026-10-25 18:33:51,876 INFO mapred.Merger: Merging 1 sorted segments
2026-10-25 18:33:51,877 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 43 bytes
2026-10-25 18:33:51,877 INFO mapred.LocalJobRunner: 2 / 2 copied.
2026-10-25 18:33:51,886 INFO Configuration.deprecation: mapred.skip.on is deprecated. Instead, use mapreduce.job.skiprecords
2026-10-25 18:33:52,327 INFO mapred.Task: Task:attempt_local1831276806_0001_r_000000_0 is done. And is in the process of committing
2026-10-25 18:33:52,329 INFO mapred.LocalJobRunner: 2 / 2 copied.
2026-10-25 18:33:52,332 INFO mapred.Task: Task attempt_local1831276806_0001_r_000000_0 is allowed to commit now
2026-10-25 18:33:52,347 INFO output.FileOutputCommitter: Saved output of task 'attempt_local1831276806_0001_r_000000_0' to hdfs://localhost:8020/user/hadoop/QuasiMonteCarlo_1792953229317_959697275/out
2026-10-25 18:33:52,348 INFO mapred.LocalJobRunner: reduce > reduce
2026-10-25 18:33:52,348 INFO mapred.Task: Task 'attempt_local1831276806_0001_r_000000_0' done.
2026-10-25 18:33:52,348 INFO mapred.Task: Final Counters for attempt_local1831276806_0001_r_000000_0: Counters: 30
        File System Counters
                FILE: Number of bytes read=282497
                FILE: Number of bytes written=1001539
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=236
                HDFS: Number of bytes written=451
```

```
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=7
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Combine input records=0
                Combine output records=0
                Reduce input groups=2
                Reduce shuffle bytes=56
                Reduce input records=4
                Reduce output records=0
                Spilled Records=4
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=351272960
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Output Format Counters
                Bytes Written=97
2026-10-25 18:33:52,348 INFO mapred.LocalJobRunner: Finishing task: attempt_local1831276806_0001_r_000000_0
2026-10-25 18:33:52,348 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-10-25 18:33:52,494 INFO mapreduce.Job: Job job_local1831276806_0001 running in uber mode : false
2026-10-25 18:33:52,495 INFO mapreduce.Job:  map 100% reduce 100%
2026-10-25 18:33:52,496 INFO mapreduce.Job: Job job_local1831276806_0001 completed successfully
2026-10-25 18:33:52,503 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=846838
                FILE: Number of bytes written=3004457
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=590
                HDFS: Number of bytes written=923
                HDFS: Number of read operations=32
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=15
                HDFS: Number of bytes read erasure-coded=0
```

```
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=2
                Map output records=4
                Map output bytes=36
                Map output materialized bytes=56
                Input split bytes=294
                Combine input records=0
                Combine output records=0
                Reduce input groups=2
                Reduce shuffle bytes=56
                Reduce input records=4
                Reduce output records=0
                Spilled Records=8
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=948436992
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=236
        File Output Format Counters
                Bytes Written=97
Job Finished in 1.626 seconds
Estimated value of Pi is 3.12000000000000000000
hadoop@ip-172-31-32-84:/opt/hadoop/share/hadoop/mapreduce$
```